TCMS software Installation Qualification – Script

QbD Quality by Design	QbD RESEARCH AND DEVELOPMENT LAB. PVT.LTD
Document No.: IQ/TCMS/1.0/00	INSTALLATION QUALIFICATION FOR TCMS APPLICATION Ver. 1.0

Document Details:

Project Name	TCMS (Training & Competency Management System)
Document Name	Installation Qualification
Prepared by	Logicology
Software Name & Version	TCMS Application Version 1.0 (package version 0.1.0)
Module Name	1. Organization Management (Units, Departments & Sections) 2. Personnel Management 3. Roles & Access Management 4. Topics, Topic Groups & Topic Folders 5. Training Content Management 6. Training Assignments & Training Calendar 7. Assessments & Question Bank 8. Qualifications & Trainer Certificates 9. Refresher Training 10. Reports 11. Notifications 12. Audit Trail 13. Password Policy
Server / Host Name	QBDSER15 (192.168.10.15), domain qbdlab3.com
Date of Preparation	07/11/2026
Applicable Guidelines	21 CFR Part 11, GAMP5, EU Annex 11

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1. Definitions.

1.1 Installation Qualification

Installation Qualification is a document that confirms and compiles all the software related information for the Installation Qualification of TCMS Application Version 1.0. This document assesses the installation site/environment, examines software instruction manuals, and validates the condition of the installed software.

1.2 Performer

The Performer is the personnel who will perform the Installation Qualification of the software. The Performer should be the personnel who is trained and has adequate knowledge about 21 CFR part 11 and the execution of Installation Qualification script.

1.3 Reviewer

The Reviewer is the person responsible for the verification of the Installation protocol.

1.4 Approver

The Approver is the person responsible for the approval of the executed Installation Qualification document.

2. Purpose:

The purpose of this Installation Qualification (IQ) document is to establish documented evidence that the TCMS (Training & Competency Management System) Application Version 1.0 has been installed according to pre-approved business specifications and requirements. This includes confirming the software's installation and compatibility with the organization's operational environment.

3. Scope:

This document applies to the IQ of the TCMS Application Version 1.0 (all modules) installed on server QBDSER15 (192.168.10.15) at QbD Research and Development Lab Pvt. Ltd. The scope includes:

- Proper installation of the software.
- Verification of installation site and environmental conditions.
- Validation of hardware and software specifications.
- Verification of installation of the runtime, database, object storage and application services.
- Confirmation of required documentation, including instruction manuals, qualification protocols, and compliance certificates.

Table 1: Scope

Document Name	INSTALLATION QUALIFICATION
Software Name & Version	TCMS Application Version 1.0

Module Name	1. Organization Management (Units, Departments & Sections) 2. Personnel Management 3. Roles & Access Management 4. Topics, Topic Groups & Topic Folders 5. Training Content Management 6. Training Assignments & Training Calendar 7. Assessments & Question Bank 8. Qualifications & Trainer Certificates 9. Refresher Training 10. Reports 11. Notifications 12. Audit Trail 13. Password Policy

3.1 System Overview

a. Description of TCMS Application Ver. 1.0

The TCMS (Training & Competency Management System) Application facilitates:

- Management of organization structure (units, departments and sections) and personnel master with role-based access.
- Creation of training topics, topic groups, topic folders and controlled training content (SOPs / documents) with version control.
- Assignment and scheduling of trainings, including training calendar and refresher training.
- Online assessments with question bank and evaluation of trainee competency.
- Qualification of personnel and issue of trainer certificates.
- System notifications and e-mail alerts.
- Reports, password policy management and comprehensive audit trails for all actions.

b. System Requirements

- Server: HPE ProLiant DL360 Gen10, Intel(R) Xeon(R) Gold 6254 CPU @ 3.10GHz, 64 GB RAM.
- Windows Server 2022 Standard (64-bit).
- Node.js 24.14.1 (64-bit).
- PostgreSQL 16 (16.6-1).
- MinIO Object Storage Server (Community Edition) – C:\minio.
- NSSM 2.24 64-bit (Windows Service Manager) – C:\nssm.
- Microsoft Edge (latest version).

c. Network Configuration

- Static IP configuration (192.168.10.15), host QBDSER15.qbdlab3.com.
- TCMS application port 3000 (HTTP) – http://192.168.10.15:3000
- PostgreSQL port 5432 (application connects via localhost).
- MinIO ports 9000 (S3 API) and 9001 (Console), bound to 127.0.0.1 (local to server).

4. Addendums, Supplements & Revisions:

Addendums, supplements, and revisions to this document are permitted. A description of the change must be entered in the Revision History at the end of this document.

If it becomes necessary to amend, supplement, or revise this document, these changes will be prepared in the format of this document, and will be reviewed and approved by those individuals specified in the last revision of the original plan or an appointed designee.

5. Installation Qualification Scripts:

Installation Scripts are applicable as per the system as per Design Specification document.

Table 2: Installation Qualification Scripts

Sr. No.	Test Title	Test Objective
1.	Server Identification	To verify server name, manufacturer and model.
2.	Server Components and Software Components.	To validate hardware and software specifications.
3.	Server Network Connection & Date and Time Format	To verify the system network connectivity. To verify the system date and time format.
4.	Runtime and Supporting Software Version Verification	To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.
5.	Windows Services Verification	To verify that TCMS components are installed as Windows services, are running and are configured to start automatically.
6.	Database Installation Verification	To verify that the TCMS application database is created and all database schema migrations are applied.
7.	Object Storage (MinIO) Verification	To verify that the MinIO object storage used for TCMS documents is accessible and the application bucket is created.
8.	Application Installation and Configuration Verification	To verify that the TCMS application files are installed and configured for the production environment.
9.	Application Accessibility Verification	To verify that the TCMS application is accessible over the network.

6. General Instructions:

The following instructions will be followed during execution of the Installation Scripts:

- The tester(s) executing each Installation Script should read the entire script before execution.
- If an exception (error) in the Installation Script or software defect is found, a discrepancy form will be completed, and the deviation added to the discrepancy log. The discrepancy number must be noted in the Actual Results column.
- The same person cannot execute the installation steps and review the obtained results.
- Screenshots shall be captured as objective evidence for each step and attached to the Executed IQ document with the corresponding Test ID, Step No. and Attachment No.
- Passwords, secret keys and other credentials shall be masked before capturing any screenshot.
- The Reviewer of the completed installation will ensure that all steps in the Installation Scripts have been executed, ensure that all executed steps are initialed and dated, check that all referenced incidents have been made, check that there are at least as many discrepancy Log entries as there are discrepancy Forms, and check that all required evidence has been documented and is attached.

7. Abbreviations, Terms and Definitions:

This section defines the abbreviations, acronyms, terms and definitions that are specific to this document.

Table 3: Abbreviations, Terms, and Definitions.

Abbreviations	Terms and Definition
TCMS	Training & Competency Management System
SOP	Standard Operating Procedure
ID	Identification
IQ	Installation Qualification
NA	Not Applicable
OS	Operating System
RAM	Random Access Memory
IP	Internet Protocol
URL	Uniform Resource Locator
HTTP	Hypertext Transfer Protocol
DB	Database
NSSM	Non-Sucking Service Manager (Windows service manager)
LGC	Logicology

Abbreviations	Terms and Definition
CFR	Code of Federal Regulation
GAMP	Good Automated Manufacturing Practices.

8. Installation Qualification Script:

Table 4: Installation Qualification Script

Test ID:	**8.1.1**
Environment:	Validation
Test title: Server Identification	
Test objective: To verify server name, manufacturer and model.	
Prerequisites and/or assumptions: Software installation must be complete.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Login to the operating system with admin credentials. • Press Win + R, type **msinfo32** and press Enter. • In System Summary, verify System Name, System Manufacturer & System Model.	1. System Name = QBDSER15 2. System Manufacturer = HPE 3. System Model = ProLiant DL360 Gen10

Test ID:	**8.1.2**
Environment:	Validation
Test title: Server Components and Software Components.	
Test objective: To validate hardware and software specifications.	
Prerequisites and/or assumptions: Software installation must be complete. Administrator user must be available.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Right Click on This PC from Desktop. Select Properties (Settings > System > About) and record the Processor, Installed Memory (RAM) & Operating System edition.	1. Processor = Intel(R) Xeon(R) Gold 6254 CPU @ 3.10GHz 3.09 GHz 2. Installed Memory (RAM) = 64 GB and above 3. Operating System = Windows Server 2022 Standard
2.	• Double click on This PC & record the Disk partitions. • Right click on each drive, select Properties & record the Capacity.	1. C:\ Capacity = 400 GB and Above 2. D:\ Capacity = 2 TB and Above
3.	• Open Control Panel > Programs and Features & record the Installed Applications.	1. Node.js 24.14.1 2. PostgreSQL 16 (16.6-1) 3. Microsoft Edge

Test ID:	**8.1.3**
Environment:	Validation
Test title: Server Network Connection & Date and Time Format	
Test objective: To verify the system network connectivity. To verify the system date and time format.	
Prerequisites and/or assumptions: Server must connect with domain (qbdlab3.com).	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Open Command Prompt as Administrator. • Execute: **hostname** • Execute: **netsh interface ipv4 show config name="Embedded FlexibleLOM 1 Port 4"**	1. IP Address = 192.168.10.15 (DHCP enabled = No, i.e. Static) 2. Subnet mask = 255.255.255.0 3. Default Gateway = 192.168.10.1
2.	• Open Control Panel > Region (**intl.cpl**) and verify the date and time format.	1. Short date format = dd-MMM-yy (e.g. 07-Nov-26) 2. Time format = HH:mm (24 Hrs.)

Test ID:	**8.1.4**
Environment:	Validation
Test title: Runtime and Supporting Software Version Verification	
Test objective: To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.	
Prerequisites and/or assumptions: Software installation must be complete. Administrator user must be available.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Open Command Prompt as Administrator. • Execute the command: **node -v**	• Node.js version = v24.14.1
2.	• In Command Prompt, navigate to **C:\Program Files\PostgreSQL\16\bin**. • Execute the command: **psql --version**	• psql (PostgreSQL) 16.6
3.	• In Command Prompt, navigate to **C:\minio\bin**. • Execute the command: **minio.exe --version**	• MinIO server version is displayed. Record the version.
4.	• In Command Prompt, navigate to **C:\nssm**. • Execute: **powershell -NoProfile -Command "(Get-Item .\nssm.exe).VersionInfo | Format-List FileName,FileDescription,ProductName,FileVersion,ProductVersion"**	1. ProductName = NSSM 64-bit 2. FileVersion = 2.24

Test ID:	**8.1.5**
Environment:	Validation
Test title: Windows Services Verification	
Test objective: To verify that TCMS components are installed as Windows services, are running and are configured to start automatically.	
Prerequisites and/or assumptions: Software installation must be complete. Administrator user must be available.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Open Command Prompt as Administrator. • For each service (**postgresql-x64-16**, **tcms-minio**, **tcms-app**) execute: **sc queryex <service>** and **sc qc <service>**	1. postgresql-x64-16: STATE = RUNNING, START_TYPE = AUTO_START 2. tcms-minio: STATE = RUNNING, START_TYPE = AUTO_START 3. tcms-app: STATE = RUNNING, START_TYPE = AUTO_START

Test ID:	**8.1.6**
Environment:	Validation
Test title: Database Installation Verification	
Test objective: To verify that the TCMS application database is created and all database schema migrations are applied.	
Prerequisites and/or assumptions: PostgreSQL service must be running. TCMS database user (tcms_app) credentials must be available.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Open Command Prompt and navigate to **C:\Program Files\PostgreSQL\16\bin**. • Execute: **psql -h localhost -U tcms_app -d tcms_prod -c "SELECT datname AS database, pg_encoding_to_char(encoding) AS encoding, datcollate AS collation FROM pg_database WHERE NOT datistemplate;"**	• Database **tcms_prod** is listed (UTF8).
2.	• Execute: **SELECT current_database();** • Execute: **SELECT COUNT(*) FROM "_prisma_migrations" WHERE finished_at IS NOT NULL;** • Execute: **SELECT migration_name FROM "_prisma_migrations" ORDER BY finished_at DESC LIMIT 1;**	1. Current database = tcms_prod 2. Count of applied migrations = 33 3. Latest migration = 20261106150000_topic_folders

Test ID:	**8.1.7**
Environment:	Validation
Test title: Object Storage (MinIO) Verification	
Test objective: To verify that the MinIO object storage used for TCMS documents is accessible and the application bucket is created.	
Prerequisites and/or assumptions: tcms-minio service must be running. MinIO administrator credentials must be available.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• On the server, open Microsoft Edge and navigate to **http://localhost:9001**. • Login with MinIO administrator credentials.	• MinIO Console login page is displayed and login is successful.
2.	• Open Object Browser / Buckets in the MinIO Console.	• Bucket **tcms-documents** is listed.

Test ID:	**8.1.8**
Environment:	Validation
Test title: Application Installation and Configuration Verification	
Test objective: To verify that the TCMS application files are installed and configured for the production environment.	
Prerequisites and/or assumptions: Software installation must be complete. Administrator user must be available.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Open File Explorer and navigate to the TCMS application installation directory **C:\tcms**. • Record the installation path.	1. Installation path = C:\tcms 2. Directory contains: **.next**, **node_modules**, **prisma**, **public**, **package.json** and **.env**.
2.	• In Command Prompt at **C:\tcms**, display the name, version and Next.js entries of **package.json** (read only).	1. "name" = "tcms" 2. "version" = "0.1.0" 3. "next" = "14.2.35"
3.	• In Command Prompt at **C:\tcms**, display **.env** (read only) with all passwords / secret keys masked.	1. NEXTAUTH_URL = http://192.168.10.15:3000 2. DATABASE_URL points to localhost:5432/tcms_prod 3. STORAGE_PROVIDER = minio 4. MINIO_BUCKET = tcms-documents

Test ID:	**8.1.9**
Environment:	Validation
Test title: Application Accessibility Verification	
Test objective: To verify that the TCMS application is accessible over the network.	
Prerequisites and/or assumptions: tcms-app service must be running. Workstation must be connected to the network with Microsoft Edge installed.	
Acceptance criteria: Actual results should match with expected results.	

Step	Test Procedure	Expected Results (Kindly refer Post IQ for actual test result)
1.	• Open Microsoft Edge. • Navigate to **http://192.168.10.15:3000** (server network address).	1. TCMS Login page is displayed. 2. Browser tab title = "TCMS — Training & Competency Management System"

9. Revisions History:

Table 5: Revision History

Version	Description of Change	Revised by
00	Initial Document as per change control No: _____ dated _____	NA

10. Approval:

Prepared by	Designation	Sign & Date
	(Logicology)	
Signature indicates that the information above is accurate and the validation activities proposed are in compliance with regulatory and corporate requirements.		

Reviewed by	Designation	Sign & Date
	(Logicology)	
	(QbD Research and Development Lab Pvt. Ltd.)	
Signature indicates that the information above is accurate and the validation activities proposed are in compliance with regulatory and corporate requirements.		

Approved by	Designation	Sign & Date
	(QbD Research and Development Lab Pvt. Ltd.)	
Signature implies agreement that the content, rationale and approach of the Validation activities are consistent with regulatory and corporate requirements.		